

DONGYEONG KIM

Computer Science Undergraduate | AI Researcher
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RESEARCH INTERESTS

Neuromorphic computing, spiking neural networks, energy-efficient AI, computer vision, object detection, brain-inspired computing, and hardware-aware machine learning.

EDUCATION

- Stony Brook University (SUNY Korea)** | *B.S. in Computer Science*Feb. 2024 - Present
- Dean's List, Spring 2024.
 - Selected coursework: Calculus I, Applied Linear Algebra, Discrete Mathematics, Foundations of Computer Science, and Programming Abstractions.

RESEARCH EXPERIENCE

- Undergraduate Research Assistant** | *Brain-Inspired Computing Lab (BCL)*Mar. 2024 - Present
- Conduct research on spiking neural networks, energy-efficient object detection, and neuromorphic computing under the direction of Prof. Yoon Seok Yang.
 - Develop and evaluate Sigma-Delta spiking object-detection models, emphasizing accuracy, latency, energy, and deployment trade-offs.
 - Implement training and evaluation workflows involving dataset preparation, temporal learning, surrogate-gradient methods, and computer-vision metrics.
 - Conduct experiments with neuromorphic computing workflows, including Intel Lava and Intel Loihi 2 inference and performance evaluation.

PUBLICATIONS

Dongyeong Kim, Gilho Kim, Myeong Jun Kim, Si Yong Lee, and Yoon Seok Yang. "Advanced Training Algorithms in Sigma-Delta Spiking YOLO for Energy-Efficient Object Detection on Neuromorphic Hardware." *IEEE Access*, 2026. doi:10.1109/ACCESS.2026.3655178

Peer-reviewed open-access article. Developed and evaluated training algorithms for Sigma-Delta spiking neural networks for energy-efficient object detection and neuromorphic deployment.

ACADEMIC SERVICE AND EMPLOYMENT

- Computer Science Department Senator** | *SUNY Korea Undergraduate Student Council*2024
- Represented Computer Science student interests in academic and student-governance discussions.
- IT Services Administrative Assistant** | *SUNY Korea Work-Study Program*Mar. - Jun. 2024
- Supported IT Services administrative and operational work for university users and internal processes.

TECHNICAL SKILLS

Programming	Python, Java, C / C++
ML and AI Research	PyTorch, deep learning, computer vision, object detection, efficient AI Spiking neural networks, neuromorphic computing, temporal learning, energy-aware evaluation
Tools	Linux, Jupyter, GitHub, Intel Lava, Intel Loihi 2